FRI Natural Language Processing Course 2025/2026
Project report

University *of Ljubljana*
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Zakonodajko: Retrieval-Augmented Question Answering over Slovenian Tax Legislation

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Abstract

Zakonodajko is a Retrieval-Augmented Generation (RAG) assistant for Slovenian tax legislation. It extracts local PISRS HTML documents, chunks them, embeds the chunks with a multilingual sentence-transformer, indexes them in FAISS, retrieves grounded context, and prompts a local instruction-tuned language model. We compare a fixed dense baseline with a Version 2 pipeline that adds article-aware legal chunking and hybrid reranking. The corrected aggregate results from the existing report show that Version 2 improves source hit@3 from 40.0% to 90.0% for the full 30-question run, while the strict 20-question prompt run reaches 95.0% source hit@3, 40.0% article hit@3, and 0.65/2 manual correctness.

Keywords

retrieval-augmented generation, Slovenian legislation, tax law, FAISS, sentence embeddings

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1. Introduction

Answering legal questions requires high accuracy and explicit source grounding, especially in tax law, where obligations depend on precise statutory conditions, definitions, deadlines, exemptions, and procedures. Large language models can produce fluent answers, but without authoritative sources they may rely on incomplete knowledge or generate unsupported claims. Our project, *Zakonodajko*, addresses this by using RAG over Slovenian tax legislation instead of relying on model memory alone. The collection contains local PISRS versions of ZDDV-1 [10], ZDoh-2 [13], ZDDPO-2 [11] and ZDavP-2 [12].

The main contribution of this project is a domain-specific RAG pipeline for Slovenian tax law that combines local statutory documents with source-grounded answer generation. By focusing on a narrow legal domain, *Zakonodajko* shows how retrieval-augmented language models can support legal question answering where correctness, explainability, and authoritative citations are essential.

1.1 Related Work

Retrieval-augmented generation was introduced as a way to combine parametric model knowledge with external retrieved documents for knowledge-intensive NLP tasks [6]. This approach is well suited to legal question answering, where answers must be supported by authoritative and up-to-date legal texts rather than generated from memory.

Legal NLP has been studied through benchmarks such as LexGLUE [1] and LegalBench [3], which evaluate legal language understanding and legal reasoning. Legal information retrieval has also been explored in COLIEE, a competition focused on legal retrieval and entailment tasks [9]. More recent work focuses specifically on legal RAG. LegalBench-RAG evaluates the retrieval step of legal RAG systems and emphasizes precise retrieval of short, relevant legal passages that can support citations [8].

Prior work also shows that RAG does not automatically eliminate hallucinations. Magesh et al. [7] found that commercial legal AI tools using RAG still produced hallucinated or incorrect legal answers. This motivates our focus on strict source grounding, conservative prompting, and evaluation of whether generated answers are actually supported by retrieved Slovenian tax-law provisions.

Compared with these works, *Zakonodajko* focuses on a narrower but practically important setting: Slovenian tax legislation. Existing legal NLP and RAG benchmarks are mostly English-centered or general-domain, while our system targets Slovenian statutory tax texts from PISRS and evaluates whether RAG can support reliable answers in this specific legal domain.

2. Methodology

Given a user question q, the system first retrieves a small set of potentially relevant legal passages from a local document collection. The retrieved passages are then provided as context to a language model, which produces a grounded answer. In this section, we describe the retrieval pipeline, document segmentation strategies, embedding models, and reranking methods used in our experiments.

2.1 Corpus Processing

The document collection consists of Slovenian tax-related legal acts and implementing rules downloaded from local HTML sources. Each document is converted into plain text while preserving metadata such as the source filename and document type. For legal documents, this metadata is important because the source act, article number, and article title often determine whether a passage is relevant to a question.

After extraction, documents are segmented into smaller retrieval units. Each resulting chunk is stored as a JSONL record containing the chunk text, source filename, chunk identifier, and available legal metadata. These chunks are then embedded and indexed with FAISS[5] for efficient nearest-neighbour search.

2.2 Chunking Strategies

We consider two document segmentation approaches.

Fixed-size chunking. The baseline approach splits documents into overlapping character windows. In our baseline configuration, each chunk contains 1200 characters with 200 characters of overlap. This strategy is simple and domain-independent, but it ignores the internal structure of legal documents.

Article-aware legal chunking. To better match the structure of legislation, we introduce article-aware chunking. This strategy detects article boundaries, article numbers, and article titles, and uses them as natural segmentation units. Each chunk is enriched with metadata such as the legal act identifier, article number, and article title. If an article is longer than the target chunk size, it is split into overlapping subchunks while preserving the article metadata.

2.3 Embedding Models

Each chunk is encoded into a dense vector representation using a multilingual sentence embedding model. We compare several embedding models with different capacity and retrieval characteristics: MiniLM-L12-v2 [22], MPNet-base-v2 [19], multilingual-e5-base [21] and BGE-M3 [2]

2.4 Dense Retrieval

The simplest retrieval method is dense nearest-neighbour search. Given a question q, we compute its embedding $\mathbf{e}_q$ and retrieve the top-k chunks whose embeddings have the highest inner product with $\mathbf{e}_q$:

$$s_{\text{dense}}(q,d) = \mathbf{e}_q^\top \mathbf{e}_d.$$

This method captures semantic similarity and is fully domain-independent. However, legal retrieval often requires exact structural cues, such as article numbers, legal act identifiers, and article titles.

2.5 Hybrid Reranking

To incorporate legal structure while retaining semantic retrieval, we use a two-stage hybrid retrieval method. First, dense FAISS search retrieves a candidate set of size K, where K is larger than the final number of returned chunks. Second, candidates are reranked using a weighted combination of semantic and symbolic signals.

For a query q and candidate chunk d, the final retrieval score is defined as

$$\begin{aligned} s(q,d) = {} & s_{\text{dense}}(q,d) + \lambda_{\text{lex}} s_{\text{lex}}(q,d) \\ & + \lambda_{\text{title}} s_{\text{title}}(q,d) + b_{\text{law}}(q,d) \\ & + b_{\text{article}}(q,d). \end{aligned}$$

The dense similarity score s_{dense} is complemented by several heuristic components. The term s_{lex} captures lexical overlap between the query and the chunk content, while s_{title} measures overlap between the query and the article title. The boost term b_{law} increases the score of chunks originating from a legal act explicitly mentioned or strongly implied in the query. Similarly, b_{article} rewards chunks whose article number is directly referenced in the query.

2.6 Single-Source and Multi-Source Retrieval

We evaluate retrieval under both single-source and multi-source settings. In the single-source setting, each question is associated with one answer-bearing legal article. In the multi-source setting, a question requires evidence from two distinct legal locations. This setting is more demanding because the retrieval system must return complementary passages rather than only the single most similar chunk.

To support multi-source retrieval, we test larger final retrieval depths such as $k = 5$ and $k = 8$. These settings allow the retriever to return evidence from multiple articles while still keeping the context small enough for downstream answer generation.

2.7 Retrieval Configurations

The evaluated retrieval configurations include dense retrieval and several hybrid variants. Dense retrieval returns the top-3 nearest chunks directly from FAISS. Hybrid variants retrieve a larger candidate pool, typically 50-100 candidates, and rerank them with lexical, title, law, and article-number features. We evaluate hybrid settings with final depths of $k = 3$, $k = 5$, and $k = 8$.

2.8 Web-searching tool

We implemented an auxiliary web-searching tool for cases where the local legal corpus does not contain sufficient evidence. The tool queries external search APIs and prioritizes authoritative Slovenian legal and administrative sources such as PISRS [18], FURS [16], GOV.SI [17], e-Uprava [15], and AJPES [14]. Retrieved results are filtered by relevance and reliability before being passed to the language model for answer generation. If no sufficiently trustworthy evidence is found, the system returns a warning instead of generating an unsupported answer.

2.9 Chat tool

2.10 Evaluation Protocol

We evaluate retrieval at top-3 using source hit@3, article hit@3, exact chunk hit@3, phrase coverage, all-phrase hit@3, and a 0–2 context relevance score. Context relevance is 2 when the expected article or all phrases are retrieved, 1 when the correct source or at least one phrase is retrieved, and 0 otherwise.

For generation, we evaluate faithfulness and answer correctness on a 0–2 scale. Faithfulness measures whether the generated answer is supported by the retrieved context, while answer correctness measures whether it covers the expected legal answer.

3. Results

3.1 Dataset

To evaluate different retrieval scenarios, we created four subsets of increasing difficulty. The *citation* subset contains explicit references to legal acts and article numbers, while the *natural* subset uses semantic formulations without direct citations, testing whether retrieval generalizes beyond exact law-and-article matching. We additionally constructed *dual_natural* and *dual_mixed* subsets, where each question requires evidence from two different legal acts, making successful retrieval dependent on multi-document coverage rather than a single highly similar result. Each subset contains 30 examples sampled with fixed random seeds for reproducibility.

3.2 Retrieval experiments

We first evaluate the retrieval stage in isolation, before introducing answer generation. This allows us to measure whether the system can recover the correct legal evidence without confounding the results with language model behaviour. We compare retrieval configurations across all four datasets.

We evaluate three design axes: chunking, embedding model, and retrieval strategy. For chunking, we compare a fixed-window baseline against article-aware legal chunks of different sizes. For embeddings, we compare multilingual MPNet [19], multilingual E5 [21], and BGE-M3 [2]. For retrieval, we compare dense top-k retrieval against title-aware hybrid retrieval.

Figure 1 reports performance grouped by embedding model. BGE-M3 [2] performs best overall, followed by multilingual E5[21] and MPNet [19]. The gap is clearest on the natural and dual-source settings, where the retriever cannot rely only on explicit article references.

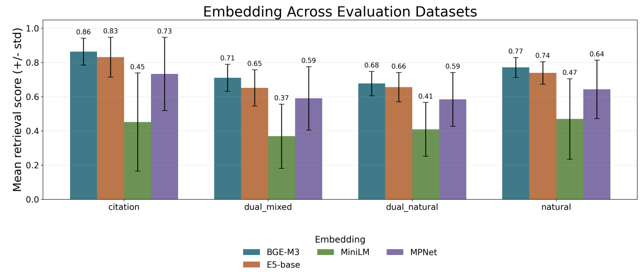

Figure 1. Retrieval performance across embedding models. Each group corresponds to one evaluation dataset, and bars report the mean retrieval score with standard deviation over chunking and retrieval configurations. BGE-M3 achieves the strongest overall performance.

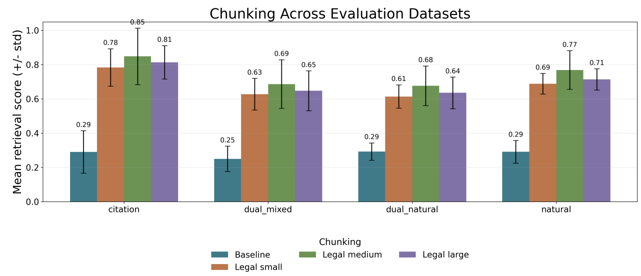

Figure 2. Retrieval performance across chunking strategies. The baseline uses fixed character windows, while legal chunking preserves article boundaries and metadata. Article-aware chunking substantially improves retrieval quality.

Figure 2 compares chunking strategies. The fixed-window baseline is consistently weaker than article-aware legal chunking. Legal-medium chunking performs best overall, suggesting that preserving article structure while keeping chunks moderately sized gives the best balance between context completeness and retrieval precision.

Figure 3 compares retrieval strategies. Dense retrieval performs worse than hybrid retrieval in all settings. Title-aware hybrid retrieval with a wider context window performs particularly well on dual-source questions, where relevant evidence must be recovered from more than one legal article.

Based on these results, we use BGE-M3 embeddings, legal-medium article-aware chunking, and title-aware hybrid retrieval in the following experiments. This configuration provides the strongest average performance while remaining robust across citation, natural, and dual-source queries.

3.3 Generation experiments

We compare two local instruction-tuned models: Mistral and GAMS9. For the final evaluation, we used a more robust dataset containing 100 questions. This final dataset was designed to provide a more reliable estimate of answer quality across a wider range of Slovenian tax-law questions.

Table 1 reports the earlier generation results obtained with

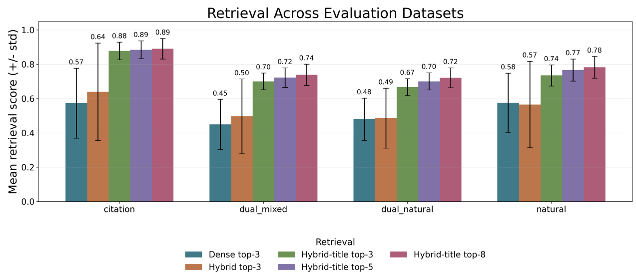

Figure 3. Retrieval performance across retrieval strategies. Hybrid retrieval improves over dense retrieval by reranking dense candidates with lexical, title, law, and article-number signals.

Mistral on smaller evaluation dataset. These results are kept for reference because they show the behaviour of the previous system version.

Table 1. Earlier generation evaluation on the old dataset using Mistral. Scores use a 0–2 scale.

Metric	Baseline prompt	Strict prompt
Context relevance	1.20	1.25
Faithfulness	1.45	1.55
Correctness	0.60	0.65

This evaluation showed that stricter prompting slightly improved correctness, but the gain was limited. This suggests that prompting alone could not fully compensate for missing or weakly relevant retrieved context. When retrieval returned only related but legally insufficient passages, the model often produced either a faithful but incomplete answer or an answer grounded in the wrong legal provision.

For the final generation evaluation, we used the improved 100-question dataset and compared Mistral[4] with GAMS9[20] under the same retrieval setting. Since both models received identical retrieved context, the retrieval metrics are the same for both systems. The comparison therefore isolates the effect of the language model used for answer generation.

Table 2. Final generation evaluation on the robust 100-question dataset. Retrieval is fixed for both models, so differences come from answer generation. Faithfulness and correctness use a 0–2 scale.

Model	Questions	Faithfulness	Correctness	Runtime
Mistral	100	1.560	0.500	19 min
GAMS9	100	1.540	0.950	32 min

Table 2 shows that GAMS9 clearly outperforms Mistral in answer correctness while maintaining almost the same faithfulness. Mistral reaches a correctness score of 0.500, while GAMS9 improves this score to 0.950. At the same time, faithfulness remains nearly unchanged: 1.560 for Mistral and 1.540 for GAMS9. This means that GAMS9 produces substantially more legally accurate answers without introducing more unsupported content.

The main trade-off is runtime. GAMS9 requires approximately 32 minutes for the full 100-question evaluation, compared with 19 minutes for Mistral. Although this makes GAMS9 slower, the large improvement in correctness makes it the better choice for final answer generation in *Zakonodajko*, where legal accuracy is more important than response speed.

3.4 Web-search Evaluation

We evaluated the web-searching component on 20 questions. For each question, the system searched the web, retrieved relevant source snippets, generated an answer, and compared it against the reference answer. The evaluation included lexical overlap with the ground truth, source quality, and an LLM-based correctness judgment.

The system achieved a mean token recall of 0.2194 and a mean token F1 score of 0.2147. These values indicate relatively low lexical overlap with the reference answers, although this metric is strict and does not fully capture semantically correct paraphrases. The expected source domain was retrieved in 30% of cases, showing that the system often found relevant sources but not always the exact expected domain. However, the LLM judge assigned a mean correctness score of 1.65 out of 5 across all 20 questions, indicating that the generated answers were often incomplete or only partially supported by the retrieved evidence.

Overall, the web-searching tool successfully retrieves relevant official sources, but the answer generation step remains weak. The results suggest that further work is needed on source selection, evidence grounding, and answer generation before the web-search component can be considered reliable for legal question answering.

Discussion and Conclusion

This report should be read as a midpoint checkpoint, not as the final evaluation of the project. The current experiments exposed an important evaluation and retrieval bias: many generated evaluation questions explicitly contain law names such as ZDDV-1, ZDoh-2, ZDDPO-2, or ZDavP-2, and the retriever uses a matching law-name signal as an additional score. This makes source correctness much higher, but it does not necessarily prove that the retriever understands the legal question or can find the exact article without this clue. Exact article matching remains unreliable, and the existing evaluation data is not complete enough to measure legal answer quality robustly.

The next stage will therefore focus on reducing this bias and building a complete human-made evaluation dataset with validated questions, exact expected sources and articles, key legal phrases, and reference answers. This should make evaluation independent of old chunk IDs and less dependent on questions that reveal the correct law in advance. In parallel, we plan to extend the RAG pipeline with additional components such as tool use, query rewriting, answer verification, or citation checking. Finally, the generation experiments will be repeated with the GAMS LLM instead of Mistral, because GAMS is a Slovenian model and should be better suited for producing grammatically and semantically correct Slovene answers.

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